



**Accelerating the Implementation of the
ASEAN Sustainable Urbanisation Strategy – Phase II**

Cities Inception Report

Phase I cities



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TABLE OF CONTENTS

City Infographics	1
Kep (Cambodia) – Solid Waste Management.....	3
Tomohon (Indonesia) – from Safety & Security to Solid Waste Management.....	5
Kaysone (Lao PDR) – from Mobility to Urban Resilience	7
Shah Alam (Malaysia) – Safety & Security	9
Mandalay (Myanmar) – Solid Waste Management	11
General Santos (The Philippines) – Mobility.....	13
Hatyai (Thailand) – Safety & Security with urban resilience	15
Sapa (Vietnam) – Mobility.....	17

CITY INFOGRAPHICS

Kep, Cambodia

Water, Waste & Sanitation



City Technical Proposal

Improvement of Solid Waste Management

Point of Attention

- none

Type of Support

- Capacity building on landfill operation and training to informal waste pickers.
- Assessment for pre-feasibility with a Capacity Gap Analysis of the Landfill.

2 Consultations

2 Women

19 Men

12 from the Government

4 from International Organizations

5 from Private Sector

Tomohon, Indonesia

Safety & Security → Waste Management



City Technical Proposal

Introduce Digital Solutions to Enhance Safety and Security in Cities

Point of Attention

- Change of priority areas due to political leadership
- Existing data related to waste

Type of Support

- Project Implementation Preparation in developing new technical proposal on solid waste management.

2 Consultations

23 Women

22 Men

37 from the Government 2 from Private Sectors

4 from International Organizations

1 from CSOs

1 from Academia

Kaysone, Lao PDR

Mobility → Urban Resilience



City Technical Proposal

Enhancing the city's drainage systems

Point of Attention

- Change of priority areas due to ongoing initiatives and funding opportunities on urban resilience.

Type of Support

- Project Implementation Preparation for Developing a new City Technical Proposal on Urban Resilience.

2 Consultations

10 Women

26 Men

27 from the Government

9 from International Organizations

Mandalay, Myanmar

Water, Waste & Sanitation



City Technical Proposal

Improvement of Solid Waste Management

Point of Attention

- Due to disruption by the earthquake, situation is being assessed to meet waste management needs.

Type of Support

- Assessment for pre-feasibility study on solid waste management after the earthquake situation.

- Consultations under preparation in coordination with development partners.
- For community-level ownership, stakeholders have been mapped.
- Community-level awareness activities have been conceptualised.

Shah Alam, Malaysia Safety & Security 	General Santos, Philippines Mobility 
City Technical Proposal Shah Alam Smart & Safe City Programme	City Technical Proposal Sustainable City Transportation Program
Point of Attention • <u>none</u>	Point of Attention • <u>None</u>
Type of Support • <u>Strategic Support</u> with the recruitment of Monitoring & Evaluation consultant related to safer cities projects.	Type of Support • <u>Application to financial opportunities</u> with the Climate City Gap Fund.
2 Consultations 15 Women  6 Men 	2 Consultations 11 Women  30 Men 
19 from the Government 	28 from the Government 
1 from CSOs 	5 from International Organizations 
1 from Academia 	4 from Private Sector  4 from CSOs 

Hatyai, Thailand Safety & Security + Urban Resilience 	Sapa, Vietnam Mobility 
City Technical Proposal Developing Safety and Security with Smart CCTV	City Technical Proposal Bus Rapid Transit System for Tourism in Sapa
Point of Attention • The new Mayor appointed in May 2025 was introduced to the ASUS Project Phase II.	Point of Attention • <u>Reform process</u> • <u>Challenges with grant support</u> for developing transport masterplan
Type of Support • <u>Strategic Support</u> with development of revised policy recommendations on disaster risk management.	Type of Support • <u>Strategic Support</u> for drafting transport related policy recommendations.
4 Consultations 11 Women  28 Men 	2 Consultations 13 Women  34 Men 
34 from the Government 	33 from the Government  2 from Private Sector 
4 from CSOs 	3 from International Organizations 
1 from Academia 	1 from CSOs  8 Academia 

KEP (CAMBODIA) – SOLID WASTE MANAGEMENT

City Background

Kep, Cambodia's smallest province, is a coastal city of about 23,000 residents, renowned for its natural attractions, beaches, and fisheries. Tourism remains the backbone of the local economy, drawing close to a million visitors annually. However, solid waste management (SWM) has emerged as the city's most pressing challenge, undermining both environmental quality and tourism appeal.

Current resources and policies

Solid waste generation in Kep ranges between 45 and 55 tonnes per day, rising sharply during peak tourist seasons. The city's Public Works, Transportation, Hygiene, Environment, and Public Order Office oversee SWM but is limited to just three staff without technical backgrounds. A 13.4-hectare landfill was constructed under an Asian Development Bank programme but remains poorly managed. Collection fees fall short of covering O&M Operational & Maintenance costs, and recycling capacity is minimal. To strengthen governance, the municipality recently formed a Waste Management Committee and a technical working group. Their purpose is to operate and maintain the landfill, but their function remains ineffective as of now due to a lack of technical training and a financial plan for cost recovery.

National policy frameworks—including the 2020–2030 National Policy on Solid Waste Management (developed and signed in 2021) and Sub-Decree 113 on Urban Waste (developed and signed in 2015)—provide strong guidance, but enforcement remains weak so far.



Figure 1 Site visit of organic waste disposal in Kep City

Priority challenges & opportunities

Kep's main challenges are limited staff capacity, underfunded operations, poor enforcement, and seasonal surges in waste. Yet, there are opportunities to build on the city's willingness to collaborate, recent institutional arrangements, and the strong policy foundation provided at the national level. Tourism provides a compelling driver to improve SWM, positioning Kep as a potential model for small coastal cities.



Figure 2 Waste segregation pilot in Kep City

Phase I to Phase II

The ASUS Project – Phase II will support Kep City in strengthening SWM through institutional capacity building, financial planning, and stakeholder engagement. The objective is to professionalise landfill operations, improve cost recovery, and mobilise community and private sector participation, thereby creating a cleaner, more resilient coastal tourism city.

During Phase I (2020-2022), the project team assisted the city in developing a city technical proposal which included the following components:

- City Solid Waste Management Plan
- Guidelines for long-term awareness campaign
- Training and demonstration of awareness campaigns
- Capacity building and installation of recycling facilities
- Set-up of an integrated recycling business model for Kep City

During Phase II, the project team identified with the city the following priority activities to advance its city technical proposal, considering the

construction of the landfill facility under the responsibility of the Ministry of Public Works and Transport of Cambodia:

- Financial gap analysis for landfill management
- Training on operations and maintenance of the landfill
- Strategy development for private and public engagement in solid waste management
- Awareness campaign for waste disposal and segregation with businesses and households

Next steps & planned support

Capacity building related to landfill operation and maintenance, and safety training to landfill staffs, informal waste pickers, and waste collection workers.

Assessment for pre-feasibility study with a Capacity Gap Analysis of the Landfill.

The ASUS Project - Phase II will recruit a SWM consultant to provide technical assistance and capacity building to Kep Municipality. The consultant will:

- **July 2025:** Carry out a financial gap analysis to assess the true costs of landfill management and recommendations for cost recovery.
- **August 2025:** Train municipal staffs and the landfill working group on the operations and maintenance of the landfill.
- **September 2025:** Develop a strategy for engaging the private sector in recycling and service delivery, aligned with Kep's needs and national policy frameworks.
- **February 2026:** Support the city in designing a public awareness campaign to increase household and business participation in proper waste disposal and segregation. with special gender equality, disability and social inclusion (GEDSI) 100 participants are expected, including 50% of women. In addition, informal waste pickers will be also invited to attend the awareness campaign.

Through this support, Kep City will be better positioned to operate its landfill sustainably and reduce the negative impacts of unmanaged waste on its coastal ecosystems and tourism industry.

ASUS Working Group Composition

City focal point	1. Mr. Khorn Vuthy, Governor of Kep City
Steering committee	Steering Committee for Waste Management <i>Chair</i> 1. Ms. Sokha Tith, Deputy Governor of Kep Province <i>Members</i> 1. Department of Public Works and Transport, 2. Department of Economic and Finance, 3. Department of Tourism, 4. Department of Environment, 5. Kep Province, 6. Kep City, and 7. Damnak Chang Er District. Working Group for Operation and Maintenance of the Landfill <i>Chair</i> 1. Mr. Buntheoun Sang, Deputy Governor of Kep City <i>Members</i> 2. members from Department Public Works and Transport, 3. Department of Environment, 4. Kep City, and 5. Damnak Chang Er District.



Figure 3 Informal waste pickers at the Kep Landfill

TOMOHON (INDONESIA) – FROM SAFETY & SECURITY TO SOLID WASTE MANAGEMENT

City Background

Tomohon, the youngest autonomous city in North Sulawesi Province, was established in 2003 and today has a population of just over 103,000 people. (49,5% women, 50,5% men). Perched at 786 meters above sea level and surrounded by two active volcanoes, the city's steep topography shapes both its development and its vulnerabilities. Tomohon has built its identity as a tourism and agricultural hub. It is part of the Likupang–Manado National Tourism Zone under the Presidential Regulation on Tourism Masterplanning (2023–2044) and hosts internationally recognised events such as the Tomohon International Flower Festival, which in 2024 generated nearly USD 5 million in economic activity. The surge in tourism and agricultural trade, however, has put growing pressure on urban services, especially solid waste management.

Current resources and policies



Figure 4 Site visit at the landfill used by Tomohon City

During the ASUS Project - Phase I, Tomohon focused on safety and security, preparing a City Technical Proposal (CTP) to expand its command centre, integrate digital safety applications, and improve incident response through systems such as call centre 112 and CCTV installations. While some elements were implemented, budget limitations prevented full rollout, leaving coverage and operational capacity constrained.

In Phase II, the city shifted its priority to Water, Waste, and Sanitation, with a strong emphasis on solid waste management. In 2023, Tomohon generated 18,747 tons of waste, but only 11.4% was formally managed. Collection services remain patchy, and the city relies on a single disposal site, *Tempat Pembuangan Akhir* (TPA) Tara-Tara, which was designed as a sanitary landfill but now operates as an open dump due to lack of equipment and staff capacity. As of 2024, operations depended on a single excavator, with the city struggling to secure the machinery and resources needed for adequate management.

Despite these constraints, promising initiatives are emerging. The Integrated Organic Waste Management and Organic Agriculture Centre/*Pusat Pengolahan Sampah Organik Terpadu dan Pertanian Organik* (PPSOT PO), led by the Family Welfare and Empowerment Movement/*Tim Penggerak Pemberdayaan dan Kesejahteraan Keluarga* (TP-PKK) in collaboration with the Environmental Agency and Agriculture Department, is piloting organic waste management in four villages and demonstrating eco-enzyme use for urban farming. Community waste banks were previously piloted with private sector partners such as *Bank Negara Indonesia* (BNI) and *PT Perusahaan Listrik Negara* (PLN) but were discontinued due to operational challenges. The city remains interested in reviving these initiatives with stronger partnerships and centralised coordination.

Tomohon's waste policies are framed by Law No. 18/2008 on Waste Management, Presidential Regulation No. 97/2017 on Household Waste (*Jakstranas*), and the Mayor's Decree No. 30/2019 on Household Waste Strategy, all of which emphasise the 3Rs (Reduce, Recycle, Reuse) and integrated systems. These frameworks set ambitious national targets, but implementation at local level remains under-resourced.

Priority challenges & opportunities

Tomohon's key challenge is the mismatch between rapidly rising waste volumes and limited capacity to manage them. Weak landfill operations, inadequate service coverage, and limited public participation all contribute to mounting environmental pressures that directly affect the tourism economy. However, the city also benefits from strong political leadership, inclusion in national tourism strategies, and a demonstrated

willingness to experiment with community-based and private-sector-supported initiatives. These create momentum for building a more circular and resilient waste management system. From a GEDSI perspective, the main issues assessed is the limited community ownership and trust in current waste management initiatives, which provide the basis for opportunities to engage more with them.

Phase I to Phase II

The aim is to move from under-resourced landfill operations and fragmented community initiatives to an integrated approach that combines improved landfill management, financial planning, public-private engagement, and community participation. This support will build on Tomohon's existing initiatives while addressing critical capacity and resource gaps.

Next steps & planned support

Project implementation preparation for developing a new City Technical Proposal on Solid Waste Management.



Figure 5 Facility for organic waste reuse in Tomohon City

The ASUS Project - Phase II will support Tomohon in developing a new CTP on circular economy approaches for solid waste management. This CTP will provide a structured roadmap for improving waste management across both upstream and downstream systems. On the upstream side, the focus will be on community engagement, plastic recycling, and organic waste reuse, building on initiatives such as the organic waste centre (PPSOT PO) and exploring ways to revive and strengthen waste banks with private sector and civic participation. On the downstream side, the CTP will address the operations and

maintenance of the Tara-Tara landfill, ensuring that it functions closer to its original sanitary design and reducing the risks associated with open dumping. By combining community-driven circular economy measures with professionalised landfill management, the CTP will create a pathway for Tomohon to transition toward a more sustainable and resilient waste system that matches its role as a growing tourism hub.

- **May 2025:** Inception workshop to assess the existing interventions, policies and resources on solid waste management. Key deliverable: Inception report.
 - **October 2025:** Diagnostic workshop to identify challenges and opportunities in solid waste management. Key deliverable: City Diagnostic Report.
 - **February 2026:** Co-design workshop to build consensus on solutions in solid waste management. Key deliverable: City technical proposal.
- May 2026:** finalisation of the City Technical Proposal.

ASUS Working Group Composition

City focal point	1. Ms. Harriet Marzan, Tomohon City Secretariate. 2. Ms. Jacqueline Mangulu, Head of City Planning and Development Agency
Steering committee	1. Representatives of: Department of Environment. 2. Department of Public Works and Spatial Planning 3. Department of Agriculture and Farming 4. Tomohon Integrated Organic Waste Management and Organic Agriculture Centre (PPSOT PO) 5. Family Empowerment and Welfare Team (TP PKK) 6. Catholic University of Tomohon 7. Sam Ratulangi University, Manado 8. Department of Connectivity and Logistic Infrastructure, Bappenas (ACSI) 9. Department of Area, Urban, and National Border, Ministry of Home Affairs (ASCN)

KAYSONE (LAO PDR) – MOBILITY TO URBAN RESILIENCE

City Background

Kaysone Phomvihane, the capital of Savannakhet Province, is a fast-growing secondary city and a strategic hub along the East–West Economic Corridor. With a population of over 120,000, the city is experiencing continuous population growth, creating both opportunities and vulnerabilities. Its

location makes it central to trade and logistics in Lao PDR, but also highly exposed to flooding and infrastructure stress linked to climate change.

Current resources and policies

During the ASUS Project - Phase I and the inception phase of Phase II, the city prioritised e-mobility and sustainable transport, preparing a City Technical Proposal (CTP) for developing a sustainable urban mobility plan. However, recent consultations, assessments, and new funding opportunities have highlighted that urban resilience has become a more urgent priority. Communities identified flood mitigation, drainage improvements, and resilient infrastructure as immediate needs.



Figure 6 Opening remarks from Kaysone officials to the ASUS inception meeting (Lao PDR)

The Department of Public Works and Transport (DPWT) is currently preparing a Wastewater Management Project targeting seven flood-prone zones, with investments in drainage upgrades and Decentralised Wastewater Treatment Systems (DEWATS) that can double as flood retention mechanisms. These initiatives align with national development strategies and contribute to multiple

SDGs, particularly SDG 6 (Clean Water), SDG 11 (Sustainable Cities), and SDG 13 (Climate Action).

Priority challenges & opportunities

Kaysone faces increasing risks from climate-induced flooding, exacerbated by urban sprawl, insufficient drainage, and limited planning capacity. Service disruptions and infrastructure stress undermine both livelihoods and economic development. Yet, the shift toward resilience offers strong opportunities: it allows the city to take a holistic approach to urban planning, integrating the enhancement of connectivity function of the regional hub while also opening doors to donor financing and technical support in climate resilience and disaster preparedness.

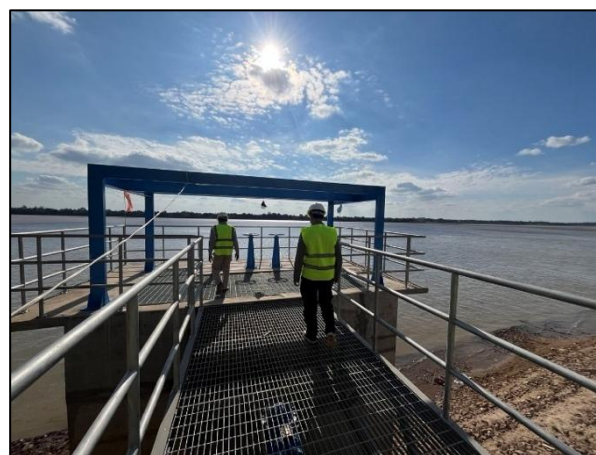


Figure 7 Site visit of water access facility in Kaysone province (Lao PDR)

Phase I to Phase II

The ASUS Project - Phase II will reposition its support to help Kaysone strengthen urban resilience as a cross-cutting priority. The project will focus on building institutional capacity for climate adaptation, improving drainage and flood protection systems, and embedding resilience into broader urban planning. By aligning local strategies with community needs and donor priorities, the ASUS Project – Phase II will position Kaysone to reduce disaster risks while creating co-benefits for public health, sustainable infrastructure, and inclusive growth. The project is supporting the city in seeking funding opportunities from the National budget programming and the international funding opportunities.



Figure 8 Urban transport station in Kaysone (Lao PDR)

Next steps & planned support

Project implementation preparation in developing a new City Technical Proposal on Urban Resilience.

The ASUS Project – Phase II will work with Kaysone Municipality to prepare a new CTP on urban resilience. The proposal will prioritise flood management, drainage upgrades, and climate-adaptive infrastructure. It envisions a climate-resilient city that ensures security for residents and confidence in investors, supporting the city's sustainable development. The ASUS Project – Phase II will also help the city connect with funding opportunities from multilateral partners and climate finance institutions, ensuring that resilience interventions can move from planning to implementation.

- **December 2024:** Inception workshop to assess the existing interventions, policies and resources on drainage systems. Key deliverable: Inception report.
- **October 2025:** Diagnostic workshop to identify challenges and opportunities in drainage systems improvement. Key deliverable: Diagnostic report.
- **February 2026:** Co-design workshop to build consensus on solutions in drainage systems improvement. Key deliverable: City technical proposal.
- **April 2026:** finalisation of the City technical proposal.

ASUS Working Group Composition

City focal point	1. Mr. Vannaeth Phaophongsavath, Deputy Head of Kaysone
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	Phomvihane City Office, as Head of Focal Points. 2. Mr. Vatthana Pongvilay, Provincial Public Works and Transportation Office, as Deputy Head of Focal Points 3. Ms. Vadsana Soumphonphakdy, Kaysone Phomvihane City Office, as a Member of the Focal Points 4. Ms. Visakha Keovilavong, Provincial Natural Resources and Environment Office, as a Member of the Focal Points 5. Mr. Chittavan Phasaysombath, Kaysone Phomvihane City Office, as a Member of the Focal Points 6. Ms. Somchai Douangsavanh, Technical Officer of the Housing and Urban Planning and Water Supply Sector, as a Member of the Focal Points
Steering committee	Representatives of: 1. Provincial Governor's Office (Savannakhet Province) 2. Kaysone Phomvihane City Administration Office 3. City Management and Services Office 4. Provincial Department of Public Works and Transport 5. District Department of Public Works and Transport 6. Provincial Department of Agriculture and Environment 7. District Department of Agriculture and Environment 8. Office of Planning 9. Department of Housing and Urban Planning, Ministry of Public Works and Transport (ASCN representative) 10. Department of Planning and Finance, Ministry of Public Works and Transport (ACSI representative)

SHAH ALAM (MALAYSIA) – SAFETY & SECURITY

City Background

Shah Alam, capital of Selangor and one of Malaysia's first planned cities, has grown into a dynamic urban hub of nearly one million residents. Spread over 303 km², it combines residential neighbourhoods, commercial centres, and traditional villages. Known for its structured planning, Shah Alam now faces the challenge of balancing growth with safety, inclusivity, and resilience.



Figure 9 Inception meeting in Shah Alam (Malaysia)

Current resources and policies

In the ASUS Project - Phase I, the city developed a City Technical Proposal (CTP) on safety and security, focusing on institutional strengthening, infrastructure improvements, and citizen engagement. Since then, Shah Alam has expanded its CCTV coverage, introduced AI traffic lights, and piloted flood-detection sensors. 74 CCTV units were installed (25 were funded by the Ministry of Economy, and 50 units were funded by Selangor State).

Its Safe City Programme, anchored in PLAN Malaysia's Safe City Guideline (2010), incorporates interventions ranging from lighting upgrades and tactile paving to community partnerships for safety audits. The Shah Alam SDG Roadmap 2030, Low Carbon Action Plan 2030, and Voluntary Local Review reinforce the city's commitment to sustainability and inclusive development.

Priority challenges & opportunities

Challenges include limited 5G coverage, uneven adoption of smart city applications, and resource constraints in scaling interventions. Yet, Shah Alam has strong governance through Majlis Bandaraya Shah Alam (MBSA), a Safe City Unit, and active partnerships with police and community associations. Opportunities lie in consolidating pilots into citywide initiatives, strengthening inclusive design, and expanding citizen uptake of safety innovations.

Phase I to Phase II

The ASUS Project – Phase II will strengthen the city's role as a leader in **urban safety and liveability**, aligned with Selangor State's Safe City and Obstacle-Free 2030 Initiative. Building on its Community Transformation Programme and existing safety initiatives, the city recognises that robust monitoring and evaluation (M&E) is essential to measure progress, engage citizens, and align outcomes with both state and ASEAN priorities. The ASUS Project - Phase II will therefore focus on enhancing Shah Alam's M&E systems through digital tools, capacity building, and peer-to-peer learning. By linking urban safety with participatory monitoring and advanced data governance, Shah Alam can demonstrate measurable results, strengthen accountability, and position itself as a model city for smart and sustainable development.



Figure 10 Site visit of key areas in Shah Alam (Malaysia)

Next steps & planned support

Strategic Support with the recruitment of a Monitoring & Evaluation consultant related to safer cities projects.

The ASUS Project - Phase II will support Shah Alam in advancing its CTP by embedding M&E as a core driver of implementation. This will include:

- Digital M&E tools: Developing participatory platforms for community-level progress tracking and co-creating customised Power BI dashboards with Microsoft to visualise the outcomes of the Community Transformation Programme.
- Capacity building: Training MBSA staff on SDG 11 benchmarking to align with global indicators, and delivering workshops on data governance, anonymisation, and compliance with the Personal Data Protection Act (PDPA) 2010.
- Third-party audits: Facilitating peer reviews and independent assessments in collaboration with ASEAN Smart Cities Network members, such as Iskandar Puteri, to enhance transparency and cross-city learning.

To operationalise this support, the ASUS Project – Phase II will recruit an M&E expert in Q4 2025 to conduct:

- **December 2025:** A gap analysis and review existing funding mechanisms.
- **January 2026:** Draft recommendations and a Key Performance Indicator framework will be developed, feeding directly into Shah Alam's CTP.

Through this stepwise approach, the ASUS Project – Phase II will ensure Shah Alam has both the technical systems and institutional capacity to track progress and deliver on its Safe City 2030 commitments.

ASUS Working Group Composition

City focal point	1. TPr. Aniza binti Haji Osman, Senior Assistant Director, Sustainable Development Unit, Department of Planning, MBSA (City Focal Point) 2. Mayor, MBSA (Chairperson)
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	3. Director, Department of Planning, MBSA 4. Department of Planning/ Secretariat for the development of MBSA Command Centre, MBSA 5. Sustainable Development Unit, Department of Planning, MBSA
Steering committee	1. Mayor, MBSA (Chairperson) 2. Secretariat for the development of MBSA Command Centre, Department of Planning, MBSA 3. Managing Director, Smart Selangor Delivery Unit 4. Department of Crime Prevention and Community Safety, Selangor Contingent Police Headquarters 5. Safe City Unit, Research and Development Division, PLANMalaysia, Ministry of Housing and Local Government (KPKT) 6. Faculty of Architecture, Planning and Surveying, Universiti Teknologi MARA 7. State Economic Planning Unit, Selangor State Secretary Office Secretariat Team: 1. Sustainable Development Unit, Department of Planning, MBSA



Figure 11 Site visit of key areas in Shah Alam (Malaysia)

MANDALAY (MYANMAR) – SOLID WASTE MANAGEMENT

City background

Mandalay City has been facing serious challenges in its solid waste management (SWM) system, especially in informal settlements, where waste collection coverage is inconsistent and infrastructure remains inadequate. Informal waste pickers operate across the city, but their contributions are marginalised due to a lack of integration with municipal systems.

Current resources & policies

To be determined during the next consultation with the identified urban stakeholders.

Priority challenges & opportunities

The recent earthquake has severely disrupted Mandalay's SWM system. Massive debris—about 2.5 million tonnes, including over 300,000 tons from collapsed homes, businesses, and religious buildings—has overwhelmed cleanup and recovery efforts.

Over 6,000 displaced households living in temporary shelters generate mixed waste such as food scraps, plastic packaging, and hygiene products. With little segregation or regular collection, waste is piling up around camps, threatening health and the environment.

Damage to key infrastructure has further interrupted waste collection and disposal, leaving neighborhoods unsanitary. Together, unmanaged debris and growing waste have increased the risk of waterborne diseases and other health hazards, especially in overcrowded sites with poor sanitation.

Women often bear increased health and care burdens when waste services are disrupted, while informal waste workers face heightened safety risks during debris handling without adequate protection or recognition. Elderly residents, persons with disabilities, and low-

income communities are particularly affected by waste accumulation, blocked access, and uneven service recovery. The project provides an opportunity to strengthen inclusive recovery by engaging community groups and women in planning processes, supporting safer practices for informal workers, prioritising vulnerable neighbourhoods in debris clearance, and embedding social inclusion considerations into capacity building and stakeholder engagement.

Phase I to Phase II

The ASUS Project - Phase I intervention aimed to create an inclusive, decentralised waste management system by formalising the role of informal waste collectors, improving municipal capacity, and implementing pilot projects in underserved areas. This was aligned with Mandalay's goal of becoming an inclusive smart city and builds on local strategic frameworks already in place.

Next steps and planned support

Assessment for pre-feasibility study

The ASUS Project - Phase II in Myanmar has been designed to advance the implementation of the previously developed Mandalay City Technical Proposal. However, the recent 7.7 magnitude earthquake that struck Myanmar on 28th March 2025 has significantly impacted ongoing operations, particularly in Mandalay.

The disaster has disrupted local coordination mechanisms, diverted municipal resources, and posed constraints on the original work plan, necessitating revisions to the implementation timeline and engagement strategy in Mandalay under the ASUS Project - Phase III. The project team will develop a comprehensive technical review report summarising key findings, gaps, and lessons learned from the ASUS Project - Phase I in Mandalay.

Additionally, a set of recommendations and updated technical notes will inform Phase II design, aligned with current priorities and contextual changes in Mandalay (assessment of rehabilitation of SWM infrastructure, effective waste collection, segregation, and disposal

mechanisms rehabilitation). In the SWM roadmap under development, special attention will be given to GEDSI, with clear opportunities to promote inclusive livelihood and social opportunities (informal sector integration, community waste banks initiatives, skill development, employment through recycling initiatives).

- **December 2025:** Training on water, sanitation and hygiene for teachers and parents at monastery schools in Mandalay

- **December 2025:** Awareness activities to promote best practices on water, sanitation and hygiene
- **February to March 2026:** Consultations to develop the solid waste management roadmap for Mandalay
- **April 2026:** Solid Waste Management Roadmap submission

ASUS Working Group composition

To be determined during the next consultation with the identified urban stakeholders.

GENERAL SANTOS (THE PHILIPPINES) – MOBILITY

City Background

General Santos City, the economic centre of Soccsksargen in southern Mindanao, is home to nearly 700,000 residents. Known as the “Tuna Capital of the Philippines,” the city plays a central role in trade, fisheries, and logistics, with urban expansion creating mounting pressures on its transport system. In the ASUS Project - Phase I, the city developed a City Technical Proposal (CTP) focused on sustainable urban mobility, which continues to frame its priorities under Phase II.



Figure 12 Inception meeting in General Santos (The Philippines)

Current resources and policies

General Santos has made steady progress in modernising its transport system. Public transport infrastructure has expanded, including traffic signalisation at 22 intersections and a command centre equipped with 52 CCTVs. Modern jeepneys are in operation, with 121 electric jeepneys and 72 Euro-IV units serving 10 active routes under the Local Public Transport Route Plan (LPTRP). The city has also advanced tricycle modernisation, issuing new franchises and initiating discussions with suppliers for electric tricycles. Financial barriers remain a concern, prompting the city to pioneer the Special Support Fund programme, one of the first nationwide efforts to provide cooperatives with financing assistance for vehicle modernisation. Policies such as Ordinance No. 36 related to the LPTRP adoption (stipulated in 2018),

Ordinance No. 37 related to the Comprehensive Land Transport Code (stipulated in 2017), and Ordinance No. 32 related to the Special Support Fund (stipulated in 2019) demonstrate strong institutional backing from the local government unit.

Priority challenges & opportunities

Key challenges include the financing gap for modern vehicles, competition between modes, and the slow introduction of e-trikes. At the same time, the city is well-placed to leverage its command centre, LPTRP, and modern fleet to strengthen sustainable and low-carbon mobility. Opportunities lie in scaling up electric vehicle adoption, securing sustainable financing for drivers and cooperatives, and institutionalising long-term management arrangements for the transport system. Women, older persons, and children are more dependent on paratransit and public transport due to lower access to private vehicles and travel needs linked to caregiving, schooling, health, and informal livelihoods. These groups rely on affordable, flexible, and nearby transport options for their daily activities.

This creates a clear opportunity to modernise and strengthen paratransit and public transport operations. Improving safety, accessibility, reliability, and service quality would disproportionately benefit these users, while also advancing transport equity, social inclusion, and access to essential services across the city.

Phase I to Phase II

The ASUS Project - Phase II will continue to support General Santos in advancing sustainable and inclusive mobility, building on the Phase I CTP. Through consultations, the objective is to accelerate the modernisation of the tricycle fleets with its electrification, given the appetite from the private sector, and the important role that tricycles play in the city's transport system.



Figure 13 Visit of the Data Command Centre in General Santos (The Philippines)

Next steps & planned support

Application to financial opportunities with the Climate City Gap Fund

The project will support advancing the CTP that consolidates interventions on modernised jeepney and tricycle fleets and explore financing pathways for e-mobility. The project team is supporting the city in applying for the City Gap Fund with assistance from the Global Covenant of Mayors and facilitating coordination with the International Finance Corporation (Apex Green Cities Programme) for loan discussions related to the electric tricycle services. Technical assistance will focus on strengthening institutional capacity, mobilising partnerships with funding partners and suppliers for electric tricycles.

- **Q4 2025:** Assistance for City Climate Gap Fund Application submission
- **February 2026:** Electric mobility city assessment report development
- **February 2026:** City workshop for report validation
- **March 2026:** Electric mobility localisation and implementation report

ASUS Working Group Composition

City focal point	1. Mr. Dominador Lagare III, City Councilor 2. Mr. Dennis Fel Matutina (Officer in charge), City Planning and Development Office 3. Engr. Mr. Rein Amoro, City Planning and Development Office
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	4. Atty. Dominador Lagare Jr., City Advisor
Steering committee	Representatives from: 1. General Santos City Mayor's Office 2. General Santos City Councilor's Office (Mr. Dominador Lagare III) 3. General Santos City Planning and Development Office (Mr. Dennis Fel Matutina, Engr. Rein Amoro) 4. General Santos City Economic Management and Cooperative Development Office (Mr. Leonard Flores, Ms. Vannessa Claire Plena) 5. General Santos City Public Safety Office (Ms. Anna Mharize Tano) 6. Tricycle Cooperatives (Ronnie Fuentes, Tito Oclarit, Josue Fernandez, Alex Dante) 7. Department of Transportation 8. Department of Human Settlements and Urban Development 9. Department of Science and Technology 10. Department of Interior and Local Government (ASCN representative) 11. Department of Planning, Economy, and Development (ACSI representative)

HATYAI (THAILAND) – SAFETY & SECURITY WITH URBAN RESILIENCE

City Background

Hatyai, Thailand's fifth-largest city and gateway to Malaysia, has a population of over 140,000 within the municipality, with the metropolitan area housing more than double that. The city is dense, commercially vibrant, and strategically located along regional transport corridors. However, its urbanisation is constrained by limited east-west connectivity and recurrent flooding.



Figure 14 Inception meeting in Hatyai (Thailand)

Current resources and policies

In the ASUS Project - Phase I, Hatyai developed a City Technical Proposal on digital safety and security, which was funded through municipal budget to expand their CCTV cameras operations and integrate it to a data command centre. The municipality approved a budget of 3.5 million USD to implement 400 CCTV cameras, aligned with the National government's Smart City Policy. Building on this, the city has advanced a climate resilience agenda, updating its Green City Action Plan 2030 and drawing on the earlier Climate Resilience Strategy developed with ACCCRN. Asian Cities Climate Change Resilience Network. Current initiatives include flood prevention planning, disaster volunteer training, community-based flood shelters, drainage studies, and the installation of broadcast towers for emergency communication. The municipality is also pursuing Smart City projects for digital services.

Priority challenges & opportunities

Hatyai faces persistent flooding, rising demand for waste and wastewater services, and vulnerabilities linked to urban density. Yet, the city has strong institutional frameworks, donor partnerships, and political will to integrate safety, climate resilience, and digital innovation. Opportunities exist to link digital safety investments with flood management and climate adaptation.

Flooding and climate-related disasters do not affect everyone in Hatyai in the same way. Women – especially female-headed households and those working in informal jobs – often carry a heavier burden during recovery and may face increased safety risks when families are displaced. Older persons and people with disabilities can struggle to receive early warnings, evacuate safely, or access temporary shelters that meet their needs. Migrant and informal communities may also miss official communications, which leaves them more exposed during emergencies.

At the moment, disaster risk management and digital safety systems in the city do not consistently account for these differences, and data is rarely broken down by gender, age, or disability. This project is an opportunity to make Hatyai's climate adaptation and flood risk planning more inclusive – by improving how early warning information reaches different groups, ensuring a wider range of voices are involved in planning, and embedding social inclusion considerations into the Climate Change Adaptation Roadmap and watershed coordination efforts.

Phase I to Phase II

The ASUS Project – Phase II in Hat Yai will align the city's safety and security agenda with its climate resilience needs. Building on Phase I work with AI-enabled CCTV and the command centre, Phase II will focus on developing a Climate Change Adaptation Roadmap that integrates disaster risk reduction, flood management, and digital monitoring systems. In linking CCTV infrastructure with adaptation strategies and in strengthening local capacity through participatory planning and training, the ASUS Project - Phase II is assisting Hat Yai in reducing damages from climate hazards while enhancing public safety with GEDSI considerations.

Next steps & planned support

Strategic Support with the development of revised policy recommendations for natural disaster risk management.

The ASUS Project - Phase II will assist Hat Yai in developing a Climate Change Adaptation Roadmap that directly connects disaster risk management with the city's broader safety and security strategy. The work will begin with a baseline assessment of Hat Yai's climate vulnerabilities, a review of past adaptation policies such as ACCCRN (2012), and an analysis of available data. This will be followed by extensive stakeholder engagement with municipal departments, law enforcement, planners, and communities to define goals and expectations.

Building on this foundation, the ASUS Project – Phase II will support the co-development of the Climate Change Adaptation Roadmap, aligning it with existing flood monitoring initiatives and the deployment of AI-enabled CCTV. The roadmap will be refined through participatory consultations, with interventions designed for flood early warning systems, basic infrastructure upgrades, environmental quality improvements, and inclusive governance. Awareness-raising campaigns will ensure community participation in adaptation and early warning processes.



Figure 15 Mapping exercise during the city consultation in Hatyai (Thailand)

In the later stages, the ASUS Project – Phase II will focus on connecting the climate change adaptation strategy to Hat Yai's safety systems, including alignment of AI CCTV data for both surveillance and flood monitoring, integration with the city's data platform, and drafting of policy recommendations for long-term disaster risk management. Through these steps, Hat Yai will strengthen its resilience to climate change while

embedding disaster preparedness into its urban safety framework.

- **December 2025 to January 2026:** Drafting of Policy recommendations for the U-Tapao area to improve flood management.
- **February 2026:** Validation Policy workshop with Hatyai and neighbouring municipalities.

ASUS Working Group Composition

City focal point	Mr. Somporn Mueangthong, Sanitation Technical Officer
Steering committee	• Office of Water Resources Region 8 • Hatyai Songkhla Hotels Association • Songkhla Irrigation Project • Songkhla Provincial Disaster Prevention and Mitigation Office • Thai SME Confederation, Songkhla Province • Mittraphap Samakkhi Foundation (Thong Sia Siang Tung)

SAPA (VIETNAM) – MOBILITY

City Background

Sapa, located in the northwest of Vietnam within the newly merged **Yen Bai Province** (formerly Lao Cai Province), sits at an altitude of 1,500–1,800 m amidst the Hoàng Liên Sơn mountains. It is one of the country's most famous highland tourism destinations, known for **Fansipan Peak** ("**the Roof of Indochina**"), its cool alpine climate, and the cultural diversity of ethnic minorities such as the Hmong, Dao, and Ha Nhi peoples.

Before July 2025, Sapa operated as a town of 16 communes and wards with more than 80,000 inhabitants. Following Vietnam's administrative reform establishing a two-tier government structure, Sapa was reorganised into **Sapa Ward**—the former urban core covering 54.63 km² with 21,700 inhabitants—and five communes. Despite this territorial adjustment, Sapa remains the **core urban and tourism centre** of the broader Sapa National Tourist Area and continues to serve as an economic engine for the mountainous northwest region.

Current resources and policies

Sapa's development has long been guided by a series of national and provincial master plans designating it as a National Tourist Area and resort city of international standard by 2030. These include the 2016 Overall Master Plan for Tourism Development of Lao Cai Province, the 2023 Overall Master Plan for Construction of Sapa National Tourist Area to 2040, and the 2024 Urban Development Programme for Sapa District (Decision 322/QĐ-UBND).

Current resources concentrate on improving infrastructure connectivity and tourism-driven urban quality, as detailed under Project No. 04-DA/TU (2020) and Plan No. 84/KH-UBND (2021) of Lao Cai Province. Major investments include:

- Development of the Sapa Airport and upgrade of the Noi Bai – Lao Cai Expressway.
- Modernisation of urban roads, public parking facilities, and smart-city pilot projects.

- Expansion of wastewater and solid-waste treatment systems, including proposals for electric public transport;
- Upgrading of telecommunication and digital infrastructure, piloted through the province's "Smart City" project (Project 08-ĐA/TU).

Sapa continues to benefit from a mix of provincial and private investment, with Lao Cai Province mobilising approximately ₫131 trillion (2021–2025) for urban and transport development—83 % from non-state capital sources

Priority challenges & opportunities

The rapid growth of tourism and the limited capacity of existing infrastructure have turned transport into Sapa's most pressing urban challenge. During peak tourist seasons, the town centre experiences chronic congestion, narrow streets are blocked by parked vehicles, and pedestrian safety is compromised. The absence of an organised public transport system means that both residents and visitors depend on private cars and motorcycles, further worsening air pollution and road safety. The rugged topography and climate-related hazards—such as landslides and heavy rains—also complicate road maintenance and traffic management. These conditions create disproportionate barriers for women, the elderly, children and persons with disabilities. Unsafe pedestrian environments, poor accessibility, and chaotic traffic flows make everyday travel more difficult and less safe for these groups, who are more reliant on walking and shared transport options.

Yet these challenges also open pathways for transformation. Sapa's compact urban core provides a manageable scale for testing low-carbon, context-sensitive mobility solutions. There is an opportunity to establish a comprehensive public transport system combining small electric shuttles, designated parking hubs, and pedestrian corridors linked to major tourist destinations. At the same time, there is a strong opportunity to embed GEDSI principles by prioritising safer sidewalks and crossings, accessible vehicle and stop design, better lighting, and more inclusive service planning, ensuring that mobility improvements

directly enhance access, safety, and comfort for vulnerable users.

Additionally, improved data collection and digital tools introduced through smart-city initiatives can enable better coordination among authorities and inform evidence-based transport planning. These factors make Sapa a suitable pilot city for demonstrating sustainable mobility in mountainous environments.

Phase I to Phase II

During the ASUS Project - Phase I, Sapa developed an **Integrated City Transport Master Plan (ICTMP)** as its City Technical Proposal. The plan outlined a structured pathway to improve mobility and reduce congestion through data-based planning, institutional strengthening, and the establishment of a sustainable public transport network. It proposed a series of outputs ranging from baseline transport surveys and the development of a transport model to the formulation of an ICTMP, the design of pilot projects, and the creation of short-, medium-, and long-term strategies for public transport development.

However, implementation was delayed due to the COVID-19 pandemic and subsequent administrative restructuring. With the 2025 reform, the scope of the city's jurisdiction was reduced from 681 km² (the former town) to 54.63 km² (the current ward).

The project team provides its support to implement the components of the City Technical Proposal from Phase I, as indicated below.

Next Step and Planned Support

Technical support for integrated transport planning in the Functional Urban Area, including Sapa Ward.

- **Q1 2026:** Sapa Transportation masterplan assessment report
- **Q1 2026:** Organisation of participatory activities in Sapa ward related to sustainable urban mobility
- **Q2 2026:** Technical proposal for a smart transportation hub

ASUS Working Group Composition

City focal point	• Yen Bai Provincial People's Committee
Steering committee	• Department of Construction and Transportation of Yen Bai province • Sapa Ward People's Committee • Economic, Infrastructure and Urban Department of New Sa Pa Ward